Given the ambiguity in the nature of assessing the “magnitude” component, we use the following criteria, tabulated in Table 3.4, to aid the assessment.

**Table 3.4.** Description of the value of magnitudes in RIAM method

| Magnitude | Description |
|---|---|
| Major positive benefit or change | Refers to significant improvements in baseline conditions and a significant reduction of stress or improvement in the baseline states of sensitive receptors. |
| Moderate positive benefit or change | Refers to significant improvements in local baseline conditions, which may lead to a moderate reduction of stress to sensitive receptors or improvement in their baseline state. |
| Minor positive benefit or change | Refers to minor positive benefit or change implies that positive changes to baseline conditions are discernible but local. These changes may lead to local and limited reduction of stress to sensitive receptors. |
| Slight positive benefit or change | Refers to slight positive benefit or change implies that changes in baseline conditions are unlikely to be detectable on-site, and thus are unlikely to cause discernible reduction of stress to sensitive receptors. |
| No change/status quo | Refers to no change/status quo implies that changes in the baseline conditions are not expected, and unlikely to cause any stress to sensitive receptors. |
| Slight negative disadvantage or change | Refers to changes in baseline conditions that are unlikely to be detectable in the field, and thus are unlikely to cause discernible stress to sensitive receptors. |
| Minor negative disadvantage or change | Refers to negative changes to baseline conditions that are discernible but local. These may also refer to changes that are approaching thresholds for established standards or guidelines. These changes may lead to a local and limited increase in stress to sensitive receptors. |
| Moderate negative disadvantage or change | Refers to significant adverse changes in local baseline conditions. These may also refer to changes that are very close to exceeding established standards or guidelines or causing significant ecological impacts. These changes may lead to a moderate increase of stress to sensitive receptors. |
| Major negative disadvantage or change | Refers to significant adverse changes in baseline conditions. These may also refer to changes that exceed established standards or guidelines or causing a complete loss of certain habitats or ecological components. These changes may lead to an unacceptable increase of stress to sensitive receptors. |

These values will then contribute to the condition’s environmental score, where:

$$Environmental\ Score\ (ES)\ =\ I * M * (P + R + C).$$

The ES attained for each condition correlates to a measure of its impact, tabulated in Table 3.5.

**Table 3.5**. List of ES range along with the degree of impact associated with each range

| Range | Impact |
|---|---|
| 116 to 180 | Major positive change/impact |
| 81 to 115 | Moderate positive change/impact |
| 37 to 80 | Minor positive change/impact |
| 7 to 36 | Slight positive impact |
| -6 to +6 | No impact / Status quo / Not applicable |

should be hoarded up to prevent target terrestrial fauna from returning to the site.

The wildlife inspection prior to tree felling and vegetation clearance will also check for any entrapped animals within the hoarded area to be cleared. This inspection will be valid for 7 days only, during which the trapped animals have to secured away from the site and clearance must be conducted. If more than 7 days has passed and site clearance is not done yet, the inspection should be carried out again.

In the event that any target fauna listed in Table 13.4 are encountered during this process, the following actions which have been developed with the consideration of reducing stress to fauna while ensuring the effectiveness of the exercise shall be taken:

i) Passive Shepherding: These are highly mobile species where passive shepherding is likely to be effective. When species in this category are encountered, personnel should allow the animal to move on its own accord. If necessary, personnel may talk loudly or make some noise by clapping their hands together to encourage the animal to move. If any individual fauna does not move on its own after sufficient time (i.e., up to one hour) has passed, the EMMP team is to make a decision on whether to call NParks' Animal Rescue Centre or ACRES for the appropriate removal of the animal.

   Should the team encounter a visibly injured animal, NParks' Animal Rescue Centre or ACRES should be called immediately for the next course of action.

   No attempt should be made by the EMMP team, workers, or other unqualified personnel at any point to handle animals on site. Handling animals without appropriate certification is illegal under the recent Wildlife Act of June 2020.

ii) Capture-and-release: Species in this group are less mobile and/or venomous, and a passive shepherding approach is deemed to be ineffective and/or unsafe. A capture-and-release approach will be needed to ensure safe relocation of these fauna from the site prior to construction. In the event that these species are encountered, NParks' Animal Rescue Centre or ACRES should be called immediately for the next course of action. Capture-and release of animals encountered should be conducted by an NParks' licensed animal management company.

For trees that are subjected to removal, it is necessary to check for the presence of fauna species before each individual tree is felled.

The ecologist shall inspect the tree for the presence of fauna, including birds, bats, arboreal mammals, and arboreal herpetofauna. The ecologist should do the following:

- Check the crown of the tree for bird nests
- Check along the trunk from the bottom up for holes in which animals could be nesting
- Scan the trunk and all the branches for animals using the tree
- Scan the ground for potential nests, eggs, or burrows

Photographs of all nests, tree holes, and burrows should be taken for record purposes.

the Indonesian bay leaf (*Syzygium polyanthum*). With the exception of the tree fern, the status of the 5 other recorded species were considered irrelevant in the study as they were cultivated remnants of historic human-dependent planting.

Other plants of conservation interest included 4 understory plants such as the Endangered grass fern (*Schizaea dichotoma*), the Vulnerable scorpion's tail (*Pentaphragma ellipticum*), pitcher (*Nepenthes ampullaria*) and the golden hairy fig (*Ficus aurata*). Multiple stands of *Pandanus* species that were potentially the Endangered *Pandanus atrocarpus* were also observed as riparian and low valley components of vegetation downstream beyond study area boundary.

### Fauna

A total of 129 fauna species (see Figure 4.6 for summary of results) from selected vertebrate and invertebrate groups were recorded in the baseline study. Various species of conservation significance were observed in the study area. Threatened species detected included the internationally Critically Endangered straw-headed bulbul (*Pycnonotus zeylanicus*), the locally Endangered changeable hawk-eagle (*Nisaetus cirrhatus*), blue-crowned hanging parrot (*Loriculus galgulus*), red junglefowl (*Gallus gallus*) and the Vulnerable variable featherlegs (*Copera vittata)*. Aquatic fauna surveys revealed that the Nationally Endangered Johnson's freshwater crab (*Irmengardia johnsoni*) was present in the vicinity of Stream 3 of current study area. This species of freshwater crab is one of the three endemic species found in Singapore.

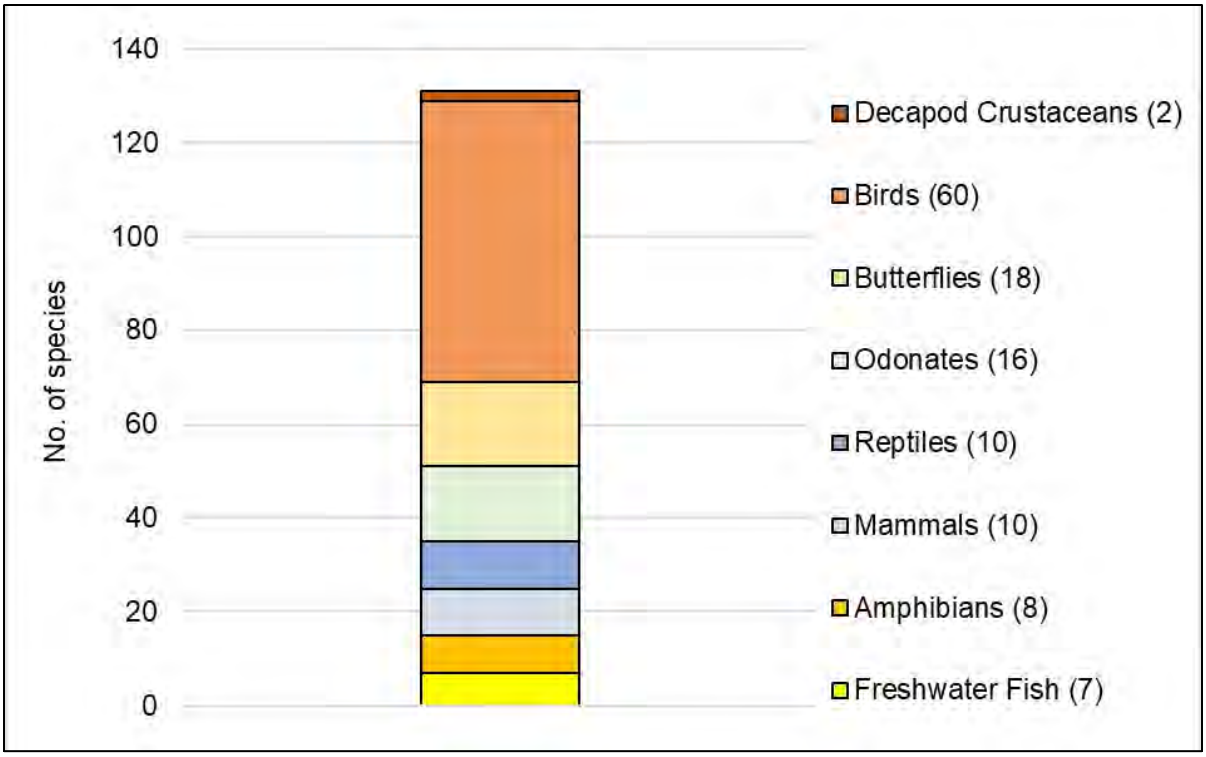


**Figure 4.6.** Summary of fauna results from previous baseline study

## 4.3.2 Review of Other Biodiversity Literature

### Flora

As Singapore is a small-city state which has undergone rapid urbanization and faces land use

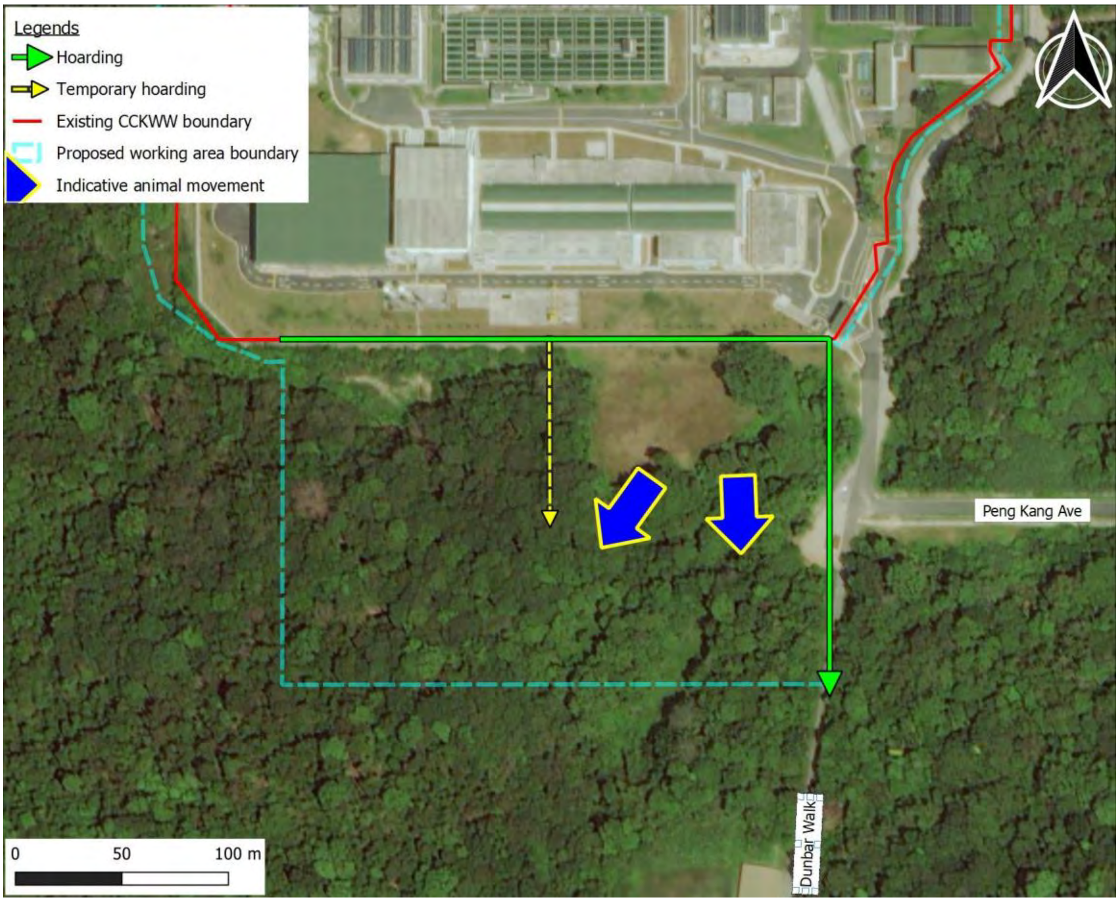


**Figure 13.2.** Directional Clearance 1

Hoarding is to be erected along the working boundaries of the project, starting with the boundary along the existing CCKWW plant and Dunbar Walk. Temporary hoarding should be erected along the yellow dotted lines. While Dunbar Walk is not used by many vehicles, speed bumps and signs informing drivers to slow down should be placed along Dunbar Walk to encourage drivers to slow down, particularly during the construction phase. Alternatively, speed limits may be implemented and enforced. Prior to any vegetation clearance, all tree holes, nests, and burrows should also be inspected for wildlife. Directional clearance should take place in the direction of the blue arrows, to encourage fauna to move to the forested areas south and west of the cleared section.

### 13.3.6 *Wildlife Response and Rescue Protocol*

Even upon the completion of wildlife shepherding works, it is highly probable that animals might be able to enter the site and get trapped, particularly burrowing or climbing animals. Whenever fauna is encountered within the working areas, all construction activities should be stopped immediately, and the Wildlife Response and Rescue Plan should be followed. Workers are to notify their supervisor, who will in turn contact the designated ecologist. The ecologist will then decide the next appropriate course of action.

All documentations of wildlife are to be captured in photographs, and a Wildlife Incident Form provided in **Appendix M** is to be filled.

**Table 13.5.** Wildlife response and rescue plan

| Particular | Within the project site | | | | | Outside project site |
|---|---|---|---|---|---|---|
| **Timeframe** | During working hours | | | | | Any time |
| **Animal type** | Highly mobile animals<br>(e.g., wild pig, feral dog, smooth-coated otters, and long-tailed macaques) | | Non-large animals | | | Any |
| | | | Venomous / poisonous (e.g., king cobra, black spitting cobra) | | Non-venomous / -poisonous (e.g., Malayan water monitor) | |
| **Animal condition** | Alive / Moving / Resting | Dead | Alive / Moving / Resting | Dead | Any | – |
| **Risk To human** | High | Low | High | Low | Low | – |
| **Response** | a. Stop work on work site<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna), who will contact NParks/ ACRES for next steps if | a. Barricade affected area<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna).<br>e. If required, Contractor to assist with transporting of | a. Stop work at affected area; if possible, barricade affected area<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna), who will contact | a. Barricade affected area<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna).<br>e. If required, Contractor to assist with transporting of | a. Stop work at affected area; if possible, barricade affected area<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna), who will contact | a. Notify NParks Animal Rescue Centre/ ACRES hotline if necessary |

| Monitoring Category | Impact | Monitoring Parameters | Monitoring Method | Location | Standards / Criteria | Time / Duration / Frequency | Reporting | Implementa-tion | Supervision |
|---|---|---|---|---|---|---|---|---|---|
| terrestrial habitats due to erosion of topsoil | | | including a Silty Imagery Detection System (SIDS) | | (Surface Water Drainage) Regulation 2007 | phase | Report | | |
| | Contamination of water resources through trade effluent discharge | • All parameters identified in EPM (Trade Effluent) Regulations for Controlled Watercourse<br>• Additional parameters: Aluminium, Conductivity, Turbidity, Total nitrogen, Total phosphorous, Total organic carbon, Ammonia, Enterococcus | • Ex-situ monitoring | • At every discharge outlet | • EPM (Trade Effluent) Regulations for Controlled Watercourse | • Once a month<br>• During entire construction phase | • Monthly Environmental Performance Report | • Contractor / CEMMP In-charge<br>• ECO | PUB/ SO |
| | Degradation of stream habitat (Construction stage monitoring) | • All parameters identified in EPM (Trade Effluent) Regulations for Controlled Watercourse | • In-situ and ex-situ monitoring | • Same locations as per baseline surveys at Stream 3 during construction phase | • EPM (Trade Effluent) Regulations for Controlled Watercourse<br>• Baseline results | • Fortnightly during construction phase | • Monthly Environmental Performance Report | • Contractor / CEMMP In-charge<br>• ECO | PUB/ SO |
| | Degradation of stream habitat (post-construction monitoring) | • All parameters identified in EPM (Trade Effluent) Regulations for Controlled Watercourse | • In-situ and ex-situ monitoring | • All streams | • EPM (Trade Effluent) Regulations<br>• Baseline results | • Monthly for 3 consecutive months after construction is complete | • Monthly Environmental Performance Report | • Contractor / CEMMP In-charge<br>• ECO | PUB/ SO |
| **Air Quality Monitoring**<br>• Minimization of human health & biodiversity impacts due to dust pollution | <u>On-site Visual and Compliance Monitoring</u> | | | | | | | | |
| | Fugitive dust emissions | • Verify implementation of dust suppression plan<br>• Watering to reduce dust emissions from exposed areas<br>• Washing bay<br>• Implementation of vehicular | • Visual monitoring<br>• Compliance check | All construction areas | • Approved dust suppression plan | • During entire construction phase | • Monthly Environmental Performance Report | • Contractor / CEMMP In-charge | PUB/ SO |